Stanley Nelavala

AI Automation Engineer (Early Career): Agents, Workflows and Security-Minded Platforms

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Huntington, IN | Open to relocate anywhere

Summary

Engineer who automates real work and keeps it secure: at a 10-person startup I automated AWS server schedules (168 to about 53 hours a week), standardized malware scanning across 9 app modules, and designed the signed service-to-service API every module uses, hardened after a security audit. I build AI agents with guardrails, prompt-injection tests and audit logs, and use Claude Code, Codex and opencode every day. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Automated weekday start and stop of dev and acceptance servers and databases (ECS, RDS): about 53 hours a week instead of 168.
- Standardized malware scanning of file uploads (GuardDuty) across 9 app modules.
- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Fixed a regular expression denial-of-service bug in a shared validation package: an 80-character hostile input took about 138 s; the new linear pattern matched the old results on 60,000 random inputs.
- Built RxDesign, the hospital design service (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine and an LLM setting; 220+ offline API tests run with fake models, so tests never call an LLM.
- Replaced offset paging with keyset paging on the catalog device list: row 150,000 of a 328,000-device dev database fell from 285 ms to 0.20 ms, with no skipped or repeated rows (tested with inserts, deletes and ties).
- Use Claude Code, Codex and opencode every day to design, write, debug and review code; tests and written acceptance checks gate every change.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR, BERTopic) that analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation.

AI projects (public code, tests)

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis* GitHub

- Multi-step agent with typed state, one constrained MCP write tool, human approval that resumes after restart, and step and token budgets; hybrid retrieval with access labels.
- 62 deterministic replay evals, prompt-injection tests, output guardrails, PII redaction, an append-only audit log; failover between two LLM adapters with a circuit breaker.

Agent-fleet build workflow | *Claude Code, opencode, DeepSeek, shell* GitHub

- A planner, manager and worker agent pipeline that delivered two public services and an ops console in 34 work packets, each with written acceptance checks and a progress log.

Customer Integration Console | *TypeScript, Next.js, Fastify, PostgreSQL, OIDC* GitHub

- Internal console with single sign-on (OIDC) in front of a Fastify and PostgreSQL API; image scanned with Trivy.

Skills

AI and automation: LangGraph, MCP tools, workflow automation, guardrails, prompt-injection tests, PII redaction, audit logs, LLM failover
AI-assisted coding: Claude Code and subagents, Codex, opencode, a DeepSeek harness; planner, manager and worker runs with acceptance checks
Languages: Python, TypeScript, Go, Rust, SQL, Java, shell
Security and cloud: Service-to-service auth, rate limiting, replay protection, OIDC, GuardDuty, Trivy, AWS (ECS, RDS, S3, SQS), Docker, GitHub Actions
Quality: pytest, Playwright, k6, OpenTelemetry, acceptance checks

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)